

ARCHANA SAMPATI

Hyderabad, India

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience in Machine Learning, NLP, and Large Language Model integration. Experienced in building Retrieval-Augmented Generation systems, semantic search applications, and text classification models. Strong expertise in Python, FAISS, prompt engineering, and deploying AI solutions on AWS.

TECHNICAL SKILLS

Programming: Python, SQL

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, LangChain

AI/ML: Machine Learning, Deep Learning, NLP, LLM Integration, Prompt Engineering, RAG

Vector Database: FAISS

Cloud: AWS

Concepts: Embeddings, Semantic Search, TF-IDF, Model Deployment, REST API Integration

PROFESSIONAL EXPERIENCE

AI Engineer - Maxton Technology Pvt Ltd | March 2024 – Present

Project: Internal Document Query Assistant (RAG-Based System)

- Designed and developed a document-based Question Answering system using Retrieval-Augmented Generation (RAG).
- Implemented document chunking, embedding generation, and vector indexing using FAISS.
- Integrated LLM APIs for context-aware response generation.
- Designed structured prompts to reduce hallucinations and improve answer accuracy.

- Optimized top-k similarity search for better semantic retrieval performance.
 - Deployed the solution on AWS infrastructure.
 - Reduced manual document search effort by 35% across teams.
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Software Engineer

Capgemini | March 2023 – March 2024

Project: Customer Feedback Analysis System

- Developed an NLP-based text classification system for analyzing customer feedback.
 - Performed text preprocessing including tokenization, stopword removal, and vectorization using TF-IDF.
 - Trained and validated a Logistic Regression model achieving 85–88% accuracy.
 - Integrated the ML model into enterprise applications using REST APIs.
 - Automated categorization of customer complaints and service queries.
 - Collaborated with cross-functional teams during requirement analysis, development, and testing phases.
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EDUCATION

B.Tech – Electronics and Communication Engineering

2018 – 2022